

Beyond Scores: Earables as Active Recovery Boosters for Longevity and Performance

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Abstract

We live in a day and age where longevity and performance are the greatest aspirations of today's society. Endurance athletes are on a quest to minimize their cardiovascular age and boosting their peak performance. Yet, the key to their ambitions lies in their recovery. Current wearables such as Oura Ring, Whoop, UltraHuman, and Garmin (Body Battery) aim at quantifying recovery by means of scores but remain passive—they monitor and occasionally advise, without intervening. With this work, we envision that *earables* are in the unique situation where they have all the attributes that it takes to move beyond passive tracking to active recovery interventions. By leveraging continuous vital signs and biometrics sensing, acoustic modulation, dynamic noise masking, and even calorie intake detection, earables could help users *actively* improve sleep, boost their nutrition and fueling, and autonomic balance. We outline this vision, discuss challenges, and call on the community to explore this paradigm shift.

CCS Concepts

• Human-centered computing → Ubiquitous computing.

Keywords

earables, recovery, intervention, longevity, wearable computing

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1 Introduction and Motivation

Longevity and peak performance have become central aspirations in modern society, fueled by advances in sports science, digital health, and a growing cultural emphasis on healthspan. For endurance athletes in particular, the goal is often not just to train harder, but

to reduce cardiovascular age and sustain top performance over longer careers. At the heart of these ambitions lies a crucial yet often underappreciated factor: *recovery*.

Recovery is an active physiological process essential for adaptation, tissue repair, hormonal balance, and overall resilience [1, 2]. Effective recovery protects athletes from accumulated fatigue, injury, and overtraining [3, 4], and is shaped by multiple interacting factors such as high-quality sleep, balanced autonomic regulation (e.g., heart rate variability), and adequate nutrition to replenish energy stores [4, 5]. In practice, it is not the training load alone that drives progress, but the body's ability to recover from it.

The past decade has seen the rise of consumer wearables like the Oura Ring, Whoop, Ultrahuman, and Garmin (Body Battery), which make recovery measurable by translating physiological data into daily readiness or recovery scores [6, 7]. These devices have helped millions better understand how sleep, stress, and activity affect their day-to-day performance. Yet despite this progress, today's wearables remain fundamentally *passive*: they monitor, summarize, and occasionally advise, but do not actively help users recover in real time [8]. For example, they might report that last night's sleep was poor, but cannot intervene to improve sleep quality during the night. They might detect signs of late-night snacking or elevated nighttime HR, but cannot gently guide the user back to healthier patterns in the moment.

We believe this limitation is not inherent to wearable technology itself, but rather reflects an opportunity for a paradigm shift: from awareness-driven tracking to *active, personalized recovery intervention*. In particular, we argue that **earables**—small, discreet devices designed for continuous wear in the ear canal—are uniquely positioned to move beyond passive monitoring. Thanks to their anatomical location and multimodal capabilities, earables can sense, decide, and act to directly influence recovery processes.

In the following sections, we describe why earables offer an ideal platform for active recovery, outline our vision of closing the loop between measurement and intervention, and discuss the challenges and research opportunities involved in making this vision a reality.

2 Why Earables? A Platform for Active Recovery

Unlike wearables such as rings and smart-bands/watches, when it comes to boosting recovery, earables offer unique opportunities



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because of their anatomical location and sensing-actuation capabilities [9]. Before we delve into the reasons that make earables the only class of wearables that can actively target and boost recovery, it is key to understand the what makes up the recipe for your green score (i.e., good recovery). The main factors that impact recovery are sleep, and heart rate (HR) - heart rate variability (HRV) balance [10, 11]. The goodness of a night sleep depends on both duration as well as its efficacy (i.e. time per sleep phase) [12]. On the other hand, HR and HRV are normally impacted by a number of factors, among which nutrition (i.e. caloric deficit/surplus) plays a key role [13–15].

By virtue of their natural placement, earables cater for:

- **Accurate vital signs tracking:** the ear allows accurate sensing of heart rate (HR)[16], breathing rate (BR) and breathing related biomarkers [17], and even blood pressure (BP)[18].
- **Acoustic modulation:** Playing calming music can reduce HR and promote parasympathetic activity [19].
- **Dynamic noise control:** Acting as active noise blockers during sleep, earables can protect deep sleep stages from disruption.
- **Eating detection:** Using chewing and swallowing sounds to detect eating episodes and potentially classify food types [20, 21]

Together, these make earables not just sensors, unlike other wearables, but true *real-time recovery mediators*.

3 Vision: From Scores to Action

Current consumer wearables provide retrospective recovery scores based on sleep, heart rate (HR), and activity metrics. While informative, these scores lack the capacity for real-time physiological modulation. Earables, with their unique anatomical position and multimodal sensing and actuation capabilities, are well-suited to implement closed-loop systems that continuously monitor and intervene to optimize recovery.

Below, we outline key use cases demonstrating how earables can leverage physiological signals and behavioral markers for active recovery support.

Real-Time Autonomic Modulation In-ear photoplethysmography (PPG) reliably captures HR and heart rate variability (HRV), critical markers of autonomic balance [16, 22]. Detecting sustained low HRV or elevated resting HR can trigger acoustic biofeedback, such as guided breathing, to promote parasympathetic activation and cardiovascular recovery [23].

Sleep Stage–Aware Acoustic Intervention Ear-EEG allows sleep stage classification with high accuracy [24, 25], enabling real-time detection of deep sleep phases. Targeted noise masking can then minimize sleep fragmentation caused by environmental disturbances [26].

Nutritional Intake Detection In-ear acoustic and inertial sensors detect chewing and swallowing, facilitating recognition of eating episodes [27]. Beyond binary detection, we envision future earables that continuously monitor caloric intake throughout the day. By integrating signal features such as chewing cadence, bite count, and swallowing acoustics, earables could estimate caloric

load, meal timing, and eating speed—allowing personalized feedback aligned with metabolic recovery goals.

Cognitive Load and Stress Monitoring Combined PPG, inertial, and acoustic sensing enable detection of stress markers such as HRV suppression and irregular breathing [16, 28, 29]. Earables can deliver brief relaxation interventions to alleviate cognitive fatigue during prolonged mental exertion [30].

Continuous Blood Pressure Estimation Pulse transit time and multi-site PPG measurements enable non-invasive blood pressure monitoring, allowing early detection of cardiovascular stress and targeted recovery interventions [18, 31].

Personalized Sleep Environment Adaptation Sensors for ambient temperature, light, and motion can infer sleep environment conditions. Earables may adjust in-ear temperature and noise masking dynamically to optimize sleep comfort and quality [32].

Hydration Status Monitoring Vascular tone changes detected via PPG can indicate hydration levels [33], enabling fluid intake reminders that support metabolic recovery [34].

Physical Activity and Posture Feedback Inertial measurement units (IMUs) track head posture and movement, allowing earables to prompt corrections for sedentary behavior or poor posture, reducing musculoskeletal strain [35].

Respiratory Pattern Monitoring and Optimization Analysis of breathing variability and patterns supports stress and recovery assessment [36], with real-time biofeedback enabling improved autonomic regulation through optimized breathing techniques [37].

Circadian Rhythm and Chronotype Integration Longitudinal data on sleep-wake cycles and physiological signals allow inference of individual circadian phases [38]. While direct light therapy is infeasible in-ear, earables can deliver context-aware auditory or somatosensory cues aligned with circadian timing to support recovery [39].

Towards Fully Integrated Closed-Loop Recovery Systems Together, these applications illustrate earables’ potential to transition from passive monitors to active recovery mediators. Integrating multimodal sensing with real-time control and user-centered actuation enables personalized, adaptive interventions aligned with intrinsic physiological rhythms. This evolution represents a paradigm shift from static analytics toward dynamic, embodied wearable systems that harmonize with the body’s natural recovery processes.

4 Challenges and Opportunities

As of today, the earable computing field pullulates with all sort of sensing approaches [40], yet it still lags behind when it comes to actuation and intervention. In fact, to achieve our vision of earables as an active recovery booster, there are a number of key challenges to overcome. In particular, it is paramount to:

- **Define concrete intervention strategies to improve sleep quality:** these could revolve around both active noise cancellation to prevent people from waking up because of loud noises, as well as inducing a calming state with some white noise or relaxing sounds playback [41].

- **Accurately tracking calories:** tracking calories precisely is arguably one of the most important aspects in the life of an endurance athlete. Whether it is for shedding weight, recover from a hard session, or even carbs-loading in view of a demanding event. Yet, doing so without the need to manually weight each and every component of a meal is still a big challenge. Earables have the unique opportunity of being able to easily detect eating episodes and, most importantly, they could pick up mastication/swallowing sounds – which can in turn be used to identify and quantify what and how much the user is eating. This can then be mapped to track the caloric intake.
- **Vital signs measurement accuracy:** Although this does not come as a surprise for people familiar with earables sensing/computing, to successfully intervene in the space of recovery, it is more important than ever to reliably estimate HR, and HRV (as proxy of the parasympathetic activity), as well as calorie estimates in everyday conditions – even under strain and heavy motion.
- **Wearability:** Once again, this does not come as a surprise to earables practitioners, but comfort and battery life for all-day and all-night use are key. These, together with lightweightness are essential feats for a device that has to be worn by athletes.

Ultimately, we believe these challenges present rich research opportunities for the EarComp community.

5 Conclusion

Earables can bridge the gap between passive tracking and active recovery, directly helping users improve sleep, nutrition, and autonomic balance. We call on researchers, designers, and engineers to explore this paradigm shift.

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